AI Placement Mentor

Intelligent Career & Placement Assistant

FOUNDATION OF ARTIFICIAL INTELLIGENCE

KEERTHANA R - 240701256
LIVYA REMI ROSI E - 240701289

Personalized preparation • Skill analysis • Placement support

Abstract

A simple AI-assisted platform for student placement preparation.

What the system does

Collects student profile and skill details
Checks technical and aptitude skills
Calculates placement readiness score
Identifies role-based skill gaps
Suggests suitable companies
Creates a personalized learning plan

What the student gets

Personal placement readiness score
Clear skill-gap report
Daily preparation schedule
Weekly MCQ tests and performance updates
Personalized company recommendations
AI chat for doubt clarification
Mock interview practice and feedback

Problem Statement

Students need different preparation plans based on their current level.

- Students are often unsure about their **placement readiness**.
- They may not know their **strengths and skill gaps** for a specific job role.
- Choosing suitable **companies based on their current skills** can be difficult.
- Students often follow a **common preparation plan** instead of a personalized one.
- Lack of **regular tests, progress tracking and feedback** makes preparation less effective.
- Students need a single platform for **learning, assessment, interview practice and placement guidance**.

Literature Survey

Existing systems cover different parts of placement preparation.

Learning Platforms

- Online Courses & Tutorials
- Skill Development
- Programming Practice
- Certification Programs
- Self-Paced Learning
- Personalized Learning Resources

Job Platforms

- Job Search & Listings
- Company Profiles
- Resume Upload
- Job Matching
- Application Tracking
- Employer Connections

Interview Platforms

- Mock Interviews
- Coding Practice
- Interview Questions
- Technical Assessments
- Communication Practice
- Interview Feedback

Research Gap

Connecting the different stages of placement preparation.

- No Single Integrated Platform
- Limited Personalized Guidance
- Lack of Dynamic Skill-Gap Analysis
- Limited Adaptive Assessments
- Separate Learning & Job Platforms
- Limited Continuous Placement Tracking

Research Objectives

The project focuses on personalized preparation.

01 Assess

- Assess Student Skills
- Evaluate Aptitude & Technical Knowledge
- Measure Communication Level
- Calculate Placement Readiness
- Identify Strengths & Weaknesses

02 Prepare

- Identify Skill Gaps
- Create Personalized Learning Plan
- Daily Practice & Tasks
- Recommend Learning Resources
- Target Role-Based Preparation

03 Improve

- Track Learning Progress
- Analyze Test Results
- Improve Weak Skills
- Update Personalized Recommendations
- Monitor Placement Readiness

Proposed Work

Main modules planned in the AI Placement Mentor.

Student Profile

- Personal details
- Skills & CGPA
- Projects & Certifications
- Target Role

Readiness Score

- Skill-based assessment
- Aptitude & communication
- Overall placement readiness

Skill Gap

- Identify missing skills
- Role-based skill comparison
- Learning suggestions

Company Match

- Match skills with company requirements
- Personalized company recommendations
- Updated based on improvement

.

Daily Planner

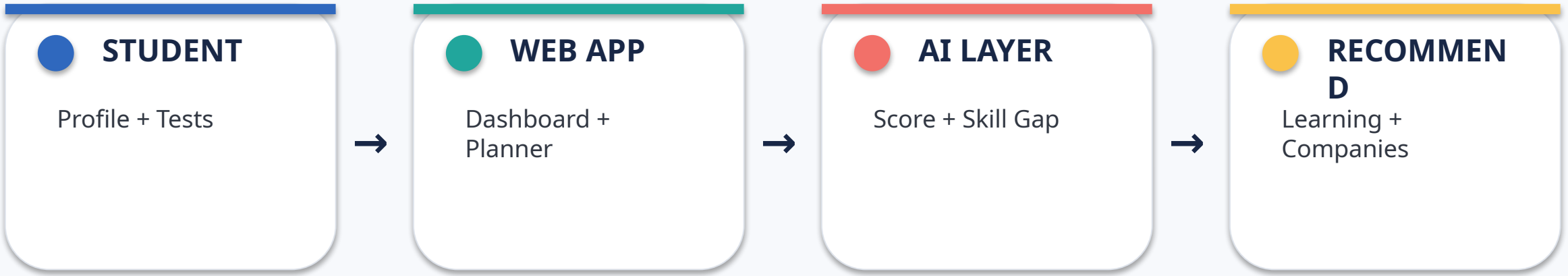
- Personalized daily schedule
- Topic-wise learning
- Practice tasks

Tests & Review

- Daily topic review
- Weekly MCQ assessment
- Performance & progress tracking .

Proposed Work – Architecture

Simple flow of the system.



Weekly results → update progress → improve the next plan

Implementation

Prototype implementation kept simple for demonstration.

- Frontend: HTML, CSS and JavaScript / React-based interface.
- Student registration and login with separate user data.
- Profile, placement score, skill-gap analysis and company matching.
- Daily planner, AI chat, daily review and weekly MCQ test.
- Mock interview video recording and feedback flow.
- Progress graph updates after assessment results.

Output & Result

The dashboard changes according to the student's performance.

Student Dashboard

- Readiness score
- Skills and gaps
- Progress graph
- Current tasks

Personalized Results

- MCQ score
- Weekly exam result
- Updated learning plan
- Interview feedback

Placement Support

- Company suggestions
- Required skills
- Preparation status
- AI doubt clearing

Conclusion

AI Placement Mentor connects assessment, preparation and placement support.

Key Outcome

- Personalized Placement Preparation
- Improved Skill Development
- Better Aptitude & Technical Performance
- Continuous Progress Tracking
- Personalized Company Recommendations
- Improved Interview Readiness

Future Scope

- AI-Based Career Guidance
- Advanced Resume Analysis
- AI Mock Interview & Feedback
- Real-Time Job & Company Updates
- Advanced Skill Recommendation
- Mobile Application Support
- Placement Prediction & Analytics

.